Worksheet
Test Prep Review 2.5 to 2.7
AP Calculus

Name_____

1.) $\lim_{x \to e} \frac{x^3 - e^3}{x - e} =$

(A) 0 (B) $3e^2$ (C) e^3 (D) Does Not Exist

2.) $\lim_{h \to 0} \frac{\sqrt{25+h} - 5}{h} =$

(A) 0 (B) 5 (C) $\frac{1}{5}$ (D) $\frac{1}{10}$ (E) $\frac{1}{25}$

3.) Given $f'(x) = \frac{1}{x}$ and $f(2) = 5$, write an equation for the line which is tangent to the graph of $f(x)$ at the point where $x = 2$.

(A) $y = \frac{1}{2}x - \frac{1}{2}$ (B) $y = \frac{1}{5}x + 5$ (C) $y = \frac{1}{2}x + 4$

(D) $y = \frac{1}{5}x - \frac{23}{5}$ (E) $y = \frac{1}{2}x + 5$

4.) Given $g(x) = 2x^5 + \frac{b}{x^2}$ where b is a constant, find the value of b if $g'(2) = 180$.

(A) 10 (B) 20 (C) –40 (D) 80 (E) none of these

5.) **Calculator active.** Which of the following is an equation of the line tangent to the graph of $f(x) = x^6 - x^4$ at the point where $f'(x) = -1$?

(A) $y = -x - 1.031$ (B) $y = -x - 0.836$ (C)) $y = -x + 0.836$

(D)) $y = -x + 0.934$ (E)) $y = -x + 1.031$

6.) $\lim_{h \to 0} \frac{3(x+h)^2 + \frac{2}{x+h} - (3x^2 + \frac{2}{x})}{h}$

(A) $x^3 + \frac{2}{x}$ (B) $3x^2 + \frac{2}{x}$ (C) $6x - \frac{2}{x^2}$

(D) $6x + \frac{2}{x^2}$ (E) nonexistent

7.) What is the slope of the line tangent to the graph of $y = 2\ln x$ at the point $x = 8$?

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) 16 (E) 4

8.) $\lim_{h \to 0} \frac{\cos\left(\frac{\pi}{6}+h\right)-\cos\left(\frac{\pi}{6}\right)}{h}$

(A) 0 (B) –1 (C) $-\frac{\sqrt{3}}{2}$ (D) $-\frac{1}{2}$ (E) nonexistent

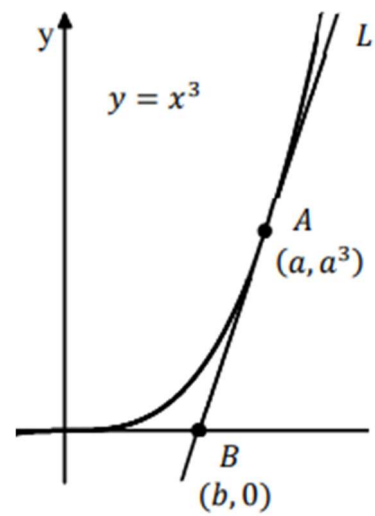
9.) **Calculator active.** Line L is tangent to the graph of $y = 2\sin x$ at the point $(k, 2\sin k)$, where $0 < k < 2\pi$. For what value of k does the line L pass through the origin?

(A) 1.571 (B) 4.493 (C) 4.712 (D) 3.141 (E) There is no such value of k

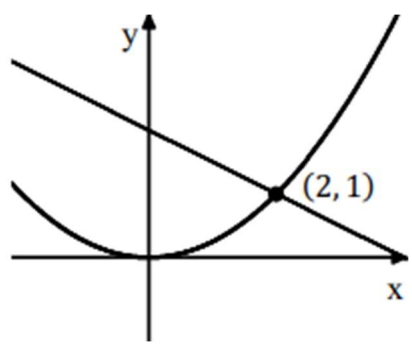
10.) In the figure to the right, line L is tangent to the graph of $y = x^3$ at point A with coordinates (a, a^3). Line L crosses the x-axis at point B, with coordinates $(b, 0)$.

a.) Find b in terms of a.

b.) Find the value of b when $a = 9$.



11.) The functions f and g are given by $f(x) = \frac{x^2}{4}$ and $g(x) = -\frac{1}{2}x + 2$.
There is a point P on the graph of f for $x \geq 0$ at which the line tangent to the graph of f is perpendicular to the graph of g. Find the coordinates of point P.



12.) $f(x) = x + a\cos x$, where a is a positive constant and $0 \leq x \leq 2\pi$. Find the x-coordinates of all points, $0 \leq x \leq 2\pi$, where the line $y = x + a$ is a tangent to the graph of $f(x)$.